

TEZPUR UNIVERSITY

End-Semester Examination, Autumn 2015

CO 504 NATURAL LANGUAGE PROCESSING

Marks : 60

Time : 2 hours

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

Submit question no. 15 within first fifteen minutes.

1. What is *IPA* in speech encoding? What are its advantages in comparison to the conventionally used scripts? 2 + 2
2. What is *morphology* in linguistics? How does morphology influence word recognition in natural language processing? 2 + 2
3. How is *n-gram* based text analysis sensitive to the training corpus? What is a *balanced corpus*? 2 + 2
4. (a) With an example, explain why part-of-speech tagging is not a trivial exercise? 3
(b) Perform PoS tagging for the following sentence- 3
I want to purchase a gift for him from the market.
5. Give examples of different forms of noun phrases and write context free grammar rules for the different forms. 2 + 3
6. Why are parsing techniques for programming languages not adequate for parsing natural languages? In this regard how is Earley's parsing technique suitable? 2 + 2

7. What aspect of syntax analysis is covered by *unification of feature structures*? 3

8. Why is a TreeBank required to implement *probabilistic context free grammar*? 2

9. Represent the following sentence using First Order Predicate Calculus- 2

Ram telephoned Sita from his office in the afternoon.

10. Identify lexical semantic relationships among the following words- 3

table (furniture), have (own), current (electricity), eat, table (information format in documents), current (flow of river), it, furniture, have (consume food).

11. How can a tagged corpus help in word sense disambiguation of the word *play* in the following sentence- 3

Ravi wrote a play that was staged on the occasion of Independence Day.

12. Give an example to illustrate the problem of reference resolution in a discourse. How can gender and number help in resolving references. 2 + 2

13. How are dialogue and monologue discourses different? 2

14. Why is *deeper semantic analysis* required in *interlingua* based machine translation than in transfer model of translation? 3

Roll No. _____

Detach this page and submit it within first fifteen minutes.

15. (I) Fill in the blanks 1 × 8

(a) Nyquist-Shannon sampling theorem is relevant in _____ (speech / text) encoding.

(b) The regular expression $/^ [Tt]he[rs]e/$ would _____ (accept / reject) the string *these*.

(c) Obtaining the word *government* from the word *govern* is an instance of _____ (derivation / inflection).

(d) The proportion of relevant documents retrieved out of all documents retrieved is _____ (recall / precision).

(e) em Grounding is a phenomenon observed in _____ (monologue / dialogue).

(f) em Grounding is a phenomenon observed in _____ (monologue / dialogue).

(g) When translation is required between one another among several languages _____ (direct / interlingua based) translation is more appropriate approach.

(h) The *fluency* of sentences produced via machine translation, can be assessed using _____ (n-grams/parallel corpora).

(II) State whether each of the following statements are true or false- 1×8

- (a) *Chair* is a synonym of *furniture*.
- (b) In the sentence “Ram has a truck” *has* is the *main* verb.
- (c) “Before Ram bought the coat, he compared the prices in several shops”. In this sentence, *he* is a cataphora.
- (d) The sentences “Ram came to office this morning. Kamal does not like sweets.” lack coherence.
- (e) In Unicode BMP is one of several planes of 64 K codes.
- (f) Polysemy may lower the precision of information retrieval.
- (g) In the sentence “Kamal said that Vimal bought himself a new shirt” *himself* refers to Kamal.
- (h) The higher the entropy of a system, the higher is the predictability of its behaviour.